

# E-Commerce Sales & Category Performance Reporting (End-to-End SQL + Excel + Power BI | YOY Commercial Analysis Project)

## Executive Summary

This project replicates a real-world **weekly Category Performance Reporting System** that I owned and delivered in my previous role as a Junior Data Analyst. The reporting solution was designed to provide commercial leadership with **regular visibility into sales performance, category contribution trends, and** performance movement across major cities in India, **with deeper profitability analysis performed at a category level in Excel.**

The analysis evaluates **weekly and monthly performance across 35 cities and 13 product categories**, comparing **2023 vs 2024** to help identify growth opportunities, category momentum, and shifts in commercial performance that inform planning and inventory decisions.

The project combines **detailed Excel-based category analysis** with an **interactive Power BI executive dashboard**, demonstrating my ability to translate raw transactional data into **clear, decision-ready insights** and build reporting systems that support high-impact commercial decision-making.

## Business Problem

Retail businesses operate in highly competitive environments where assortment, pricing strategy and regional performance insights are critical.

Retail leadership teams require:

- Detailed category-level performance tracking for commercial decisions
- A consolidated dashboard to monitor overall sales health, trends, and growth
- Visibility across categories, cities, and customers without manual reporting

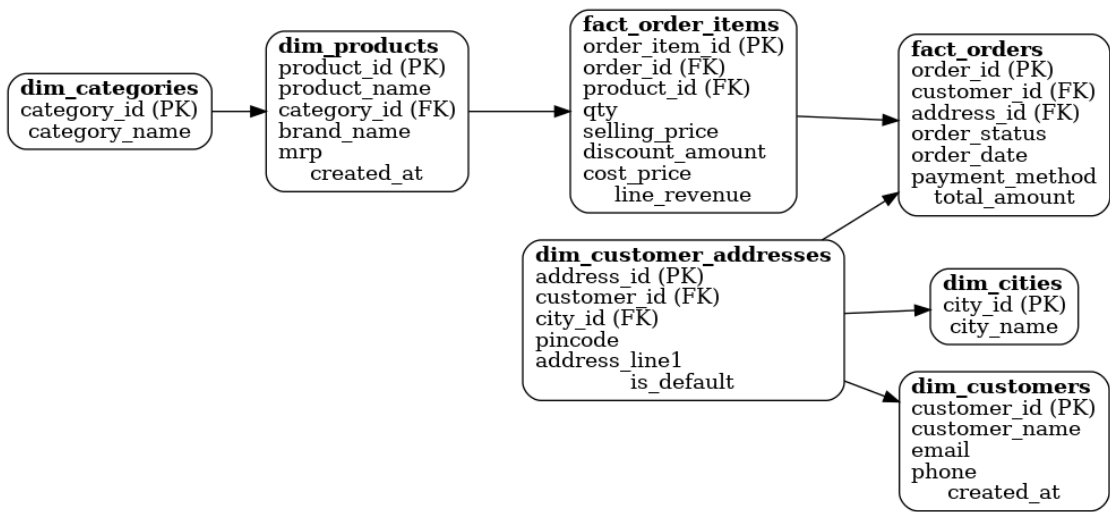
This project was designed to mirror that requirement by separating:

- **Deep-dive performance analysis**
- **High-level performance visibility**

# Data Architecture & ERD

The project follows a **star-schema** design similar to real ecommerce database modeling.

| Table                  | Description                        |
|------------------------|------------------------------------|
| dim_categories         | Product category reference         |
| dim_cities             | City master                        |
| dim_products           | Product details & category mapping |
| dim_customers          | Customer profile                   |
| dim_customer_addresses | Address mapping to city            |
| fact_orders            | Order-level commercial information |
| fact_order_items       | Transactional item-level details   |



To maintain confidentiality and avoid using any proprietary information, I created a synthetic dataset that mimics realistic ecommerce sales behavior, volume variation and seasonality.

## Excel Analysis: Category Performance (Analytical Layer)

The Excel component focuses on **structured, commercial analysis**, similar to a weekly performance report prepared for leadership teams.

### Analysis Performed in Excel

- Year-over-year comparison of **Sales Quantity, Net Revenue, Cost, and Profit**
- Category-level performance evaluation
- Identification of:

- Top-performing categories
- Declining or underperforming categories
- Contribution analysis to understand revenue concentration
- City-level aggregation to highlight key markets

Excel was used because it allows:

- Controlled calculations
- Tabular comparison
- Clear auditability of metrics
- Easy validation of commercial logic

This layer answers “**why**” performance is changing at a category and regional level.

### **Power BI Dashboard: Sales Overview (Executive Layer)**

The Power BI dashboard was built to provide a **single-page executive overview** of overall business performance, using the same underlying sales concepts but presented at a higher level.

#### **Key Metrics Displayed**

- Total Sales (2023)
- Total Sales (2024)
- Year-over-Year Sales Growth %
- Total Orders
- Average Order Value (AOV)

These KPIs give leadership an **instant snapshot** of business performance and direction.

## How Excel and Power BI Work Together (Important)

This project intentionally separates responsibilities:

| Tool     | Purpose                                               |
|----------|-------------------------------------------------------|
| Excel    | Detailed category, city, and YOY performance analysis |
| Power BI | High-level sales monitoring and executive visibility  |

In a real-world setup:

- **Excel** supports commercial reviews, planning, and investigation
- **Power BI** supports ongoing performance tracking and leadership reporting

Together, they form a **complete reporting ecosystem**, not two disconnected analyses.

## Key Business Insights Enabled

- Sales declined year-over-year, requiring deeper category review
- A small number of categories drive the majority of revenue
- Revenue is concentrated in specific cities and customers
- Order volume and AOV provide additional context beyond topline sales alone
- Category-level Excel analysis helps explain trends observed in the Power BI dashboard

## SQL Transformation & Metrics

### 1. Base CTE: `order_items_enriched`

The purpose of this CTE is to prepare a **clean, analysis-ready dataset** by joining all required fact and dimension tables into a single view.

In this stage, I:

- Begin with the transactional fact table **fact\_order\_items**, which contains one row per product per order.
- Join supporting dimensional tables to enrich the dataset:
  - **dim\_products** → adds `product_name` and `category_id`
  - **dim\_categories** → adds `category_name`
  - **fact\_orders** → provides `order_date` to enable time-based aggregation
  - **dim\_customer\_addresses** → links each order to a location
  - **dim\_cities** → adds the final `city_name`

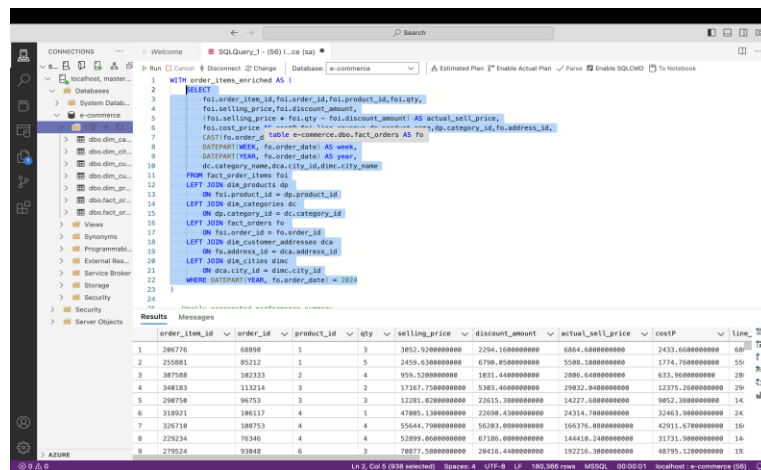
## Key calculated fields

| Field                  | Logic                                 | Purpose                                   |
|------------------------|---------------------------------------|-------------------------------------------|
| actual_sell_price      | selling_price * qty – discount amount | Net revenue after discount                |
| costP                  | cost_price                            | Cost associated with that line item       |
| week, year, order_date | extracted using DATEPART()            | Enables weekly and yearly trend reporting |

This CTE produces a **rich joined dataset** that contains:

- Category
- City
- Order timing (week and year)
- Revenue & cost at the order-item level

It serves as the **foundation** for downstream aggregation, dashboarding and YOY comparison work.



The screenshot shows a SQL Server Enterprise Manager interface. The top pane displays a SQL query for a CTE named 'order\_items\_enriched'. The query joins 'fact\_orders' with 'dim\_products', 'dim\_categories', 'dim\_customer\_addresses', and 'dim\_cities'. It calculates 'actual\_sell\_price' as 'selling\_price \* qty - discount\_amount' and includes 'cost\_price' and date parts for 'year' and 'week'. The bottom pane shows the results of the query, a table with columns: order\_item\_id, order\_id, product\_id, qty, selling\_price, discount\_amount, actual\_sell\_price, costP, week, and year. The results table contains 9 rows of data.

| order_item_id | order_id | product_id | qty | selling_price      | discount_amount    | actual_sell_price   | costP              | week | year |
|---------------|----------|------------|-----|--------------------|--------------------|---------------------|--------------------|------|------|
| 1             | 286776   | 68888      | 1   | 3852.920000000000  | 2294.100000000000  | 6864.600000000000   | 2433.600000000000  | 68   | 2024 |
| 2             | 255881   | 85212      | 1   | 2459.630000000000  | 6798.850000000000  | 5380.180000000000   | 1774.700000000000  | 55   | 2024 |
| 3             | 387588   | 182333     | 2   | 959.520000000000   | 1831.440000000000  | 2800.640000000000   | 633.900000000000   | 28   | 2024 |
| 4             | 348103   | 113214     | 3   | 17187.750000000000 | 5383.400000000000  | 23812.040000000000  | 12375.200000000000 | 29   | 2024 |
| 5             | 298759   | 96753      | 3   | 12281.020000000000 | 22615.380000000000 | 14227.680000000000  | 9852.380000000000  | 14   | 2024 |
| 6             | 318921   | 186117     | 4   | 47885.130000000000 | 22698.430000000000 | 24314.780000000000  | 32463.380000000000 | 24   | 2024 |
| 7             | 326710   | 188753     | 4   | 55644.790000000000 | 56283.000000000000 | 166376.000000000000 | 42911.670000000000 | 16   | 2024 |
| 8             | 229234   | 78340      | 4   | 52899.000000000000 | 67286.000000000000 | 144418.240000000000 | 32731.900000000000 | 14   | 2024 |
| 9             | 279524   | 92848      | 6   | 78877.500000000000 | 28435.400000000000 | 292226.300000000000 | 48795.120000000000 | 19   | 2024 |

## 2. Weekly Aggregated Performance Summary

Based on the unified CTE dataset, the final query performs an aggregation at **Category × City × Week** level.

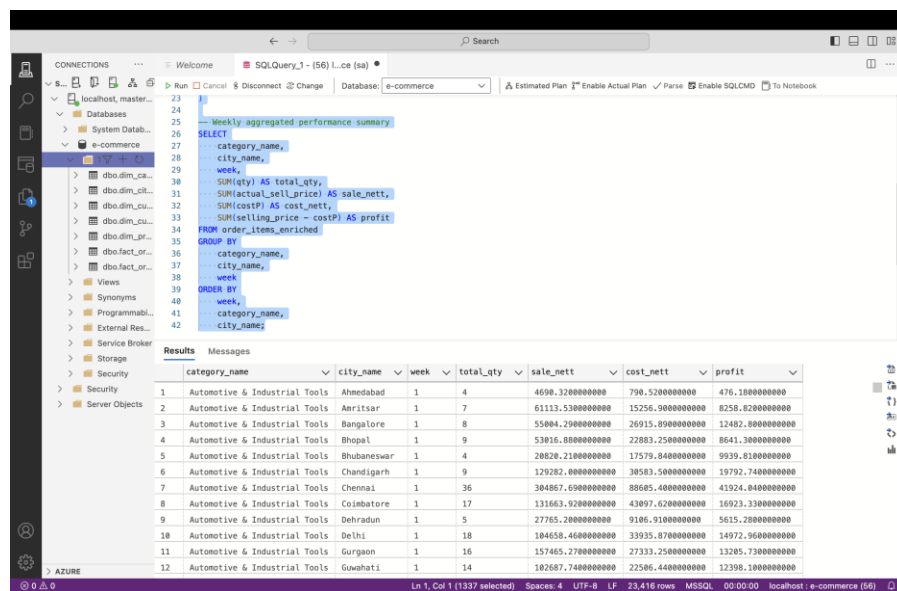
In this stage, I:

- Group by
  - category\_name
  - city\_name

- Week
- **Aggregate key performance metrics**

| Metric                     | Logic     | Purpose                      |
|----------------------------|-----------|------------------------------|
| SUM(qty)                   | total_qty | Total units sold             |
| SUM(actual_sell_price)     | sale_net  | Total revenue after discount |
| SUM(costP)                 | cost_net  | Total cost of goods sold     |
| SUM(selling_price - costP) | profit    | Profit contribution          |

The result of this query is a **weekly performance table** with the below output



The screenshot shows a SQL Server Enterprise Manager interface. The left pane displays the database structure, including databases, tables, views, and synonyms. The right pane shows a SQL query titled 'Weekly aggregated performance summary'. The query is a SELECT statement that aggregates data from the 'order\_items\_enriched' table, grouped by category\_name, city\_name, and week. The results are displayed in a table with columns: category\_name, city\_name, week, total\_qty, sale\_net, cost\_net, and profit. The results show data for 12 rows, all from the 'Automotive & Industrial Tools' category, across various cities and weeks.

| category_name                 | city_name   | week | total_qty | sale_net          | cost_net         | profit           |
|-------------------------------|-------------|------|-----------|-------------------|------------------|------------------|
| Automotive & Industrial Tools | Ahmedabad   | 1    | 4         | 4698.3200000000   | 798.5200000000   | 476.1800000000   |
| Automotive & Industrial Tools | Amritsar    | 1    | 7         | 61113.5300000000  | 15256.9000000000 | 8258.8200000000  |
| Automotive & Industrial Tools | Bangalore   | 1    | 8         | 55804.2900000000  | 26915.8900000000 | 12482.8000000000 |
| Automotive & Industrial Tools | Bhopal      | 1    | 9         | 53816.8800000000  | 22893.2500000000 | 8641.3800000000  |
| Automotive & Industrial Tools | Bhubaneswar | 1    | 4         | 28828.2100000000  | 17579.8400000000 | 9939.8100000000  |
| Automotive & Industrial Tools | Chandigarh  | 1    | 9         | 129282.8000000000 | 38583.5800000000 | 19792.7400000000 |
| Automotive & Industrial Tools | Chennai     | 1    | 36        | 384867.6900000000 | 88685.4000000000 | 41924.8400000000 |
| Automotive & Industrial Tools | Coimbatore  | 1    | 17        | 131663.9200000000 | 43897.6200000000 | 16923.3300000000 |
| Automotive & Industrial Tools | Dehradun    | 1    | 5         | 27765.2800000000  | 9186.9100000000  | 5615.2800000000  |
| Automotive & Industrial Tools | Delhi       | 1    | 18        | 184658.4600000000 | 33935.8700000000 | 14972.9600000000 |
| Automotive & Industrial Tools | Gurgaon     | 1    | 16        | 157465.2700000000 | 27333.2500000000 | 13285.7300000000 |
| Automotive & Industrial Tools | Guwahati    | 1    | 14        | 182687.7400000000 | 22586.4400000000 | 12398.1800000000 |

This is the exact structure I export to Excel to do my analysis.

## Skills Demonstrated

- Commercial performance analysis
- YOY sales reporting
- Excel-based category and contribution analysis
- Power BI dashboard design & KPI storytelling
- Translating detailed analysis into executive-level insights
- Understanding of real-world reporting workflows

## Dashboard Output & Supporting Analysis Performed

The main goal of the project was to conduct a detailed, multi-dimensional performance review to identify key trends, top performers, and areas for improvement.

- **Year-over-Year (YOY) Performance Analysis:** The core objective is to compare sales and profit performance metrics (**Quantity, Net Revenue, Net Cost, and Profit**) between 2024 and 2023.
- **Performance by Category:** The report classifies and ranks products by **Category** to determine which groups are driving the most value, notably via an **ABC Analysis** (A, B, C classifications).
- **Time-Series Trend Analysis:** Performance is tracked on a **weekly** basis to identify short-term trends, seasonality, or the impact of specific events across the entire business and within specific categories.
- **Geographical Performance Review:** The data is broken down by **City** and aggregated into **Areas/Regions** (e.g., North, South, East, West) to evaluate regional success and compare different markets.

## Business Insights (from the excel report)

| Insight                                                                      | Business Impact                                            |
|------------------------------------------------------------------------------|------------------------------------------------------------|
| <b>Mobiles &amp; Tablets and Electronics contribute ~80% of revenue</b>      | Commercial dependency risk and major strategic focus area  |
| <b>Bangalore is the top performing city with highest profit contribution</b> | Prioritize this market for promotions & premium assortment |
| <b>Footwear and Beauty categories declining YOY</b>                          | Opportunity for pricing or assortment optimization         |
| <b>ABC analysis shows 3 categories drive 85% of sales</b>                    | Inventory prioritization and marketing allocation          |

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